UNIVERSITY OF BUEA
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FACULTY OF SCIENCE
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DEPARTMENT OF COMPUTER SCIENCE

COURSE OUTLINE FOR DSC608 (Information Retrieval) – Second Semester 2023,2024
COURSE TITLE: Information Retrieval
CREDIT VALUE: 6
NAMES OF COURSE INSTRUCTORS Dr Denis L. Nkweteyim
OFFICE HOURS: _____

WEEK	SLOT		TOPICS	ACTIVITY		
	Day	Time		L	T	P
1	1		Introduction to Information Retrieval I: Structured vs Unstructured Data; What is Information Retrieval (IR); Ad-hoc Retrieval Task; Classifying IR Systems (based on scale)	√		
	2		Introduction to Information Retrieval II: Several application areas; Big Issues in IR (Relevance,Evaluation,Emphasis on users and their information needs); Search Engine Intro; Big issues in search engine design	√		
2	3		Information Retrieval System (Search Engine) Architecture I: Basic building blocks	√		
	4		Information Retrieval System (Search Engine) Architecture II: Details of building blocks: Text Acquisition, Text Transformation, Index Creation, User Interaction, Ranking, Evaluation	√		
3	5		Crawls and Feeds I: Deciding What to Search; Crawling the Web; Crawling Documents and Email; Document Feeds	√		
	6		Crawls and Feeds II: The Conversion Problem; Storing the Documents; Detecting Duplicates; Removing Noise	√		
4	7		Processing Text (Text Transformation) Ia: From Words to Terms, Text Statistics: Zipf's Law, Vocabulary Growth, Estimating Result Set Size,Estimating Collection Size	√		
	8		Processing Text (Text Transformation) Ib: Document Parsing: Overview,Tokenization,Stopping,Stemming,Phrases and N-grams	√		
5	9		Processing Text (Text Transformation) Ic: Document Structure and Markup, Link Analysis, Information Extraction	√		
	10		Review		√	
6	11		Ranking with Indexes I: Overview; Models of Ranking, Inverted Indexes	√		
	12		Ranking with Indexes II: Index Construction, Query Processing	√		
7	13		Tutorials and Revision		√	
	14		CA Test		√	
8	15		Queries and Interfaces Ia: Information Needs and Queries, Query Transformation and Refinement: Stopping and Stemming Revisited,Spell Checking and Suggestions	√		
	16		Queries and Interfaces Ib: Query Expansion,Relevance Feedback, Context and Personalization	√		
9	17		Queries and Interfaces II: Showing the Results: Result Pages and Snippets, Advertising and Search, Clustering the Results, Cross-Language Search	√		
	18		Retrieval Models: Overview of Retrieval Models: Boolean Retrieval, The Vector Space Model, Probabilistic Models	√		

10	19		Review		√	
	20		Evaluating Search Engines I Why Evaluate? The Evaluation Corpus. Logging	√		
11	21		Review		√	
	22		Evaluating Search Engines IIa Effectiveness Metrics: Recall and Precision,Averaging and Interpolation,Focusing on the Top Documents,Using Preferencest	√		
12	23		Evaluating Search Engines II Efficiency Metrics: Training, Testing, and Statistics, Significance Tests, Setting Parameter Values,Online Testing	√		
	24		Filtering and Recommendation: Document Filtering; Collaborative Filtering			
13	25		Revision		√	
	26		Revision		√	

Course Master: Denis L. Nkweteyim

Head of Department

NOTES: *L= Lecture, T = Tutorial, P= Practicals*

RECOMMENDED TEXTS

1. Croft, W. B., Metzler, D., & Strohman, T. (2010). *Search engines: Information retrieval in practice* (Vol. 520, pp. 131-141). Reading: Addison-Wesley.
2. Manning, C. D. (2008). *Introduction to information retrieval*. Syngress Publishing.

OTHER SOURCES: